CAPSTONE PROJECT REPORT

Implementation of Order-to-Cash (O2C) Process

in SAP Sales & Distribution (SD)

Submitted By

Purab Jana

Roll Number:2305956

Program: SAP SD Functional Consultant Training

Submission Date: April 21, 2026

1. Introduction

Operating across departments without a shared digital backbone breeds miscommunication, delays, and revenue leakage. SAP addresses this by unifying sales, finance, procurement, and supply chain into a single real-time platform where every transaction is traceable end to end.

Within SAP, the Sales and Distribution (SD) module manages the full customer interaction lifecycle — pricing, order management, warehouse dispatch, invoicing, and revenue recognition — while staying tightly connected with MM (Materials Management) for stock updates and FI (Financial Accounting) for revenue postings.

The Order-to-Cash (O2C) process converts a customer’s purchase intent into realized, posted revenue. When this cycle is slow or fragmented, the impact is direct: delayed shipments, billing disputes, and cash flow gaps. A well-implemented O2C in SAP replaces these pain points with an automated, audit-ready workflow.

2. Problem Statement

A mid-sized company processing hundreds of daily sales orders without an ERP faces predictable, recurring failures:

- Duplicate or incorrect order entries cause over-shipments, short-deliveries, and expensive return logistics.
- Warehouse teams rely on printed pick lists — outdated by the time goods reach the floor — leading to stale-stock dispatches.
- Invoices are raised days after delivery because billing has no automated trigger, stretching Days Sales Outstanding (DSO) and tightening cash flow.
- Credit exposure is invisible at order entry; sales reps commit to over-limit customers, creating bad debt risk.
- Revenue is estimated at month-end rather than recorded at billing, introducing audit vulnerabilities.
- No document trail exists linking invoice to original order — compliance and dispute resolution become guesswork.

3. Project Objectives

- Configure a complete SAP organizational structure for a fictitious company as the O2C implementation context.
- Execute the full O2C workflow — from customer onboarding through payment clearing — in SAP SD.
- Demonstrate live cross-module integration: SD for transactions, MM for inventory, FI for accounting.
- Document all critical T-codes and master data objects with their specific business impact.
- Validate system controls: ATP check, credit management, pricing determination, and auto-accounting.
- Identify practical future enhancements using Fiori, AI analytics, and CRM integration.

4. Organizational Structure — Arjun Industrial Supplies Pvt. Ltd.

A fictitious company has been configured in SAP to serve as the O2C implementation context. All organizational elements follow SAP best practices for a domestic Indian distribution business:

Organizational Element	Configured Value
Company Name	Arjun Industrial Supplies Pvt. Ltd.
Company Code	AIS1

Organizational Element	Configured Value
Sales Organization	AS10 — Arjun Sales India
Distribution Channel	20 — Dealer Network
Division	10 — Industrial Equipment
Plant	AP01 — Pune Manufacturing Plant
Shipping Point	ASP1 — Pune Dispatch Hub
Storage Location	SL01 — Central Warehouse, Pune
Credit Control Area	AIS1 — Domestic Credit Area

Table 1: Organizational Structure — Arjun Industrial Supplies Pvt. Ltd.

5. SAP Modules Used

Three modules operate in a synchronized chain across the O2C lifecycle:

- SAP SD (Sales & Distribution): Primary module — owns customer master data, pricing, sales orders, deliveries, and billing from start to finish.
- SAP MM (Materials Management): Activated at the delivery stage — posts stock reductions when goods are dispatched, ensuring inventory figures match physical reality.
- SAP FI (Financial Accounting): Triggered at billing — auto-generates accounting entries (debit AR, credit Revenue) and clears open items when payment arrives. No manual journal entries required.

6. End-to-End O2C Process

6.1 Customer Master (XD01) & Material Master (MM01)

Customer master (XD01) stores three data views — General Data (address, contacts), Company Code Data (reconciliation account, payment terms), and Sales Area Data (pricing, shipping priority). Every downstream SD and FI posting for that customer is governed by this record. Material master (MM01) defines the product: base unit, tax classification, item category group, and ATP check group. Skipping either setup means no transaction can proceed — SAP enforces this dependency.

6.2 Inquiry and Quotation (VA11 / VA21)

An Inquiry (VA11) records initial customer interest without commitment. The sales team converts it into a Quotation (VA21) — a time-bound offer with locked pricing and validity dates. When the sales order is raised with reference to this quotation, SAP carries pricing forward automatically, eliminating re-entry errors.

6.3 Sales Order Creation (VA01)

VA01 formalizes the commercial commitment. SAP captures sold-to/ship-to parties, materials, quantities, delivery date, and payment terms. Pricing determination pulls from pre-configured condition records automatically. A credit check runs simultaneously — orders that breach the customer’s credit limit are blocked before they reach the warehouse.

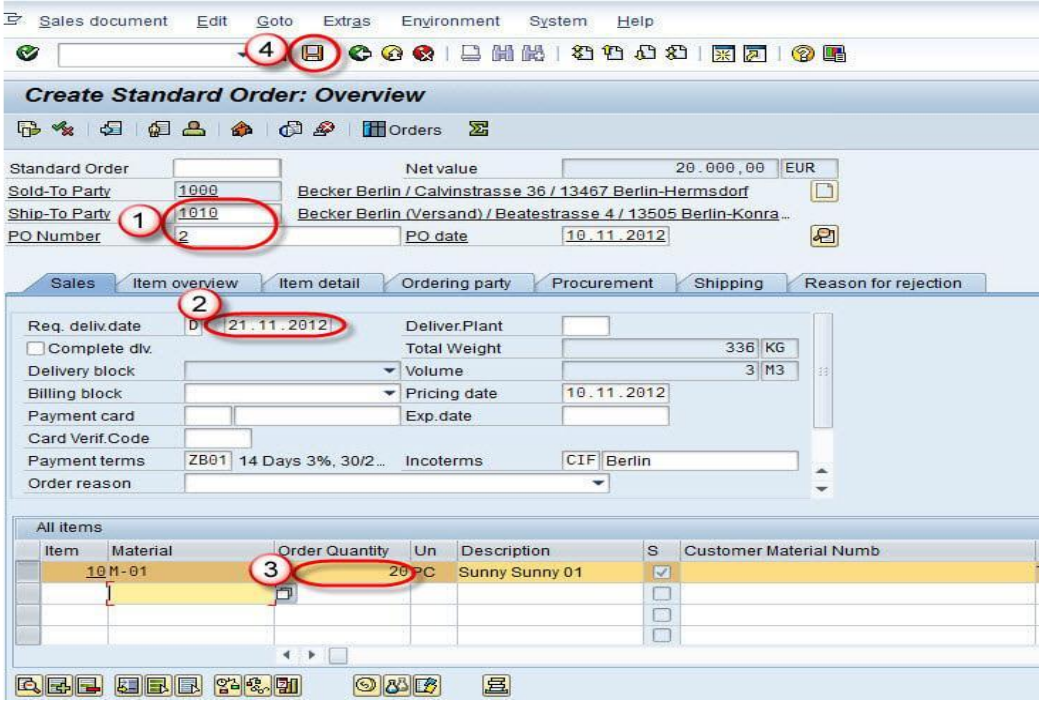

Figure 1: Sales Order Creation — T-Code VA01

6.4 Availability Check (ATP)

On saving the sales order, SAP runs an ATP (Available-to-Promise) check — evaluating unrestricted stock, existing outbound commitments, pending inbound receipts, and planned production orders. If stock is insufficient, SAP proposes the earliest confirmed delivery date rather than allowing an empty promise to the customer.

6.5 Outbound Delivery (VL01N)

VL01N creates the delivery document — the formal handoff from commercial planning to warehouse execution. It specifies materials, quantities, storage location, and dispatch date. The relevant stock is locked at this point, preventing it from being reallocated to other orders.

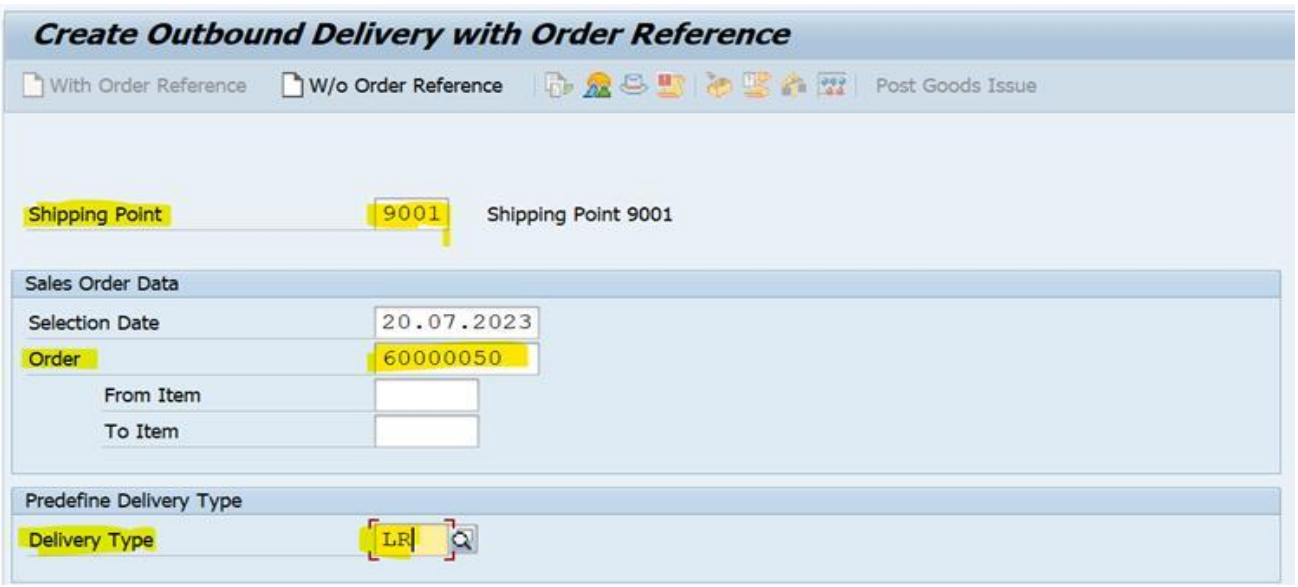

Figure 2: Outbound Delivery Process — T-Code VL01N

6.6 Picking, Packing & Post Goods Issue (VL02N)

Warehouse staff pick goods based on the delivery document and confirm completion in VL02N. Post Goods Issue (PGI) is the pivotal system event: MM reduces stock, a material document is created, and COGS is posted in FI simultaneously. The goods legally transfer to the customer at PGI — it is irreversible without a formal returns process.

6.7 Billing and Invoice Generation (VF01)

PGI completion triggers billing via VF01. SAP copies pricing from the sales order, applies output tax, and generates a commercial invoice (type F2). On save, FI auto-posts the accounting entry: Revenue account credited, customer AR account debited. Credit memos (G2) and debit memos (L2) handle post-billing corrections.

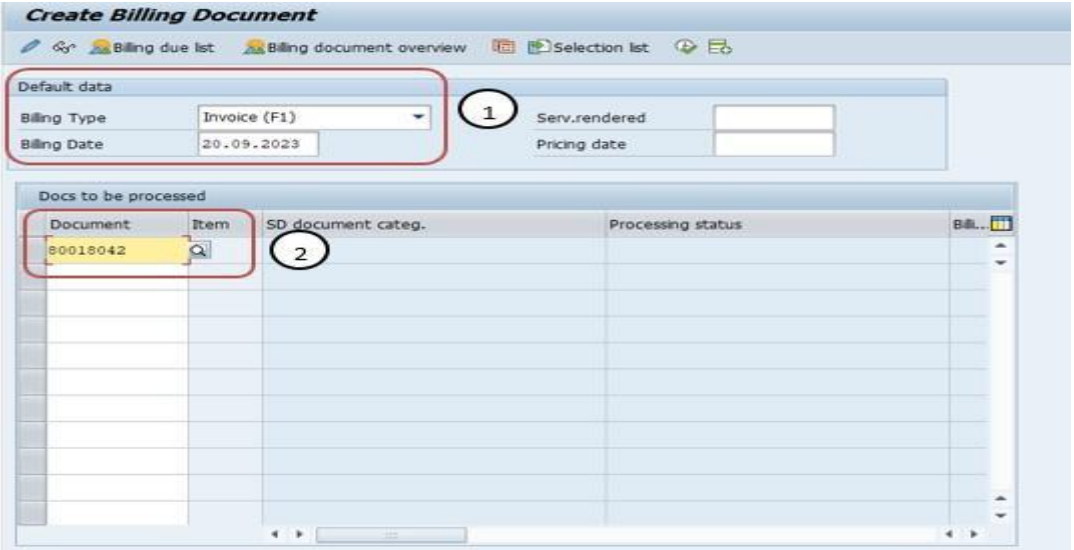

Figure 3: Billing and Invoice Generation — T-Code VF01

6.8 Payment Receipt and Clearing (F-28 / F-32)

Incoming customer payment is posted using F-28, then matched against the open invoice using F-32. This clears the Accounts Receivable balance and closes the O2C cycle. The entire revenue loop — from order to cash — is financially sealed with a traceable document chain.

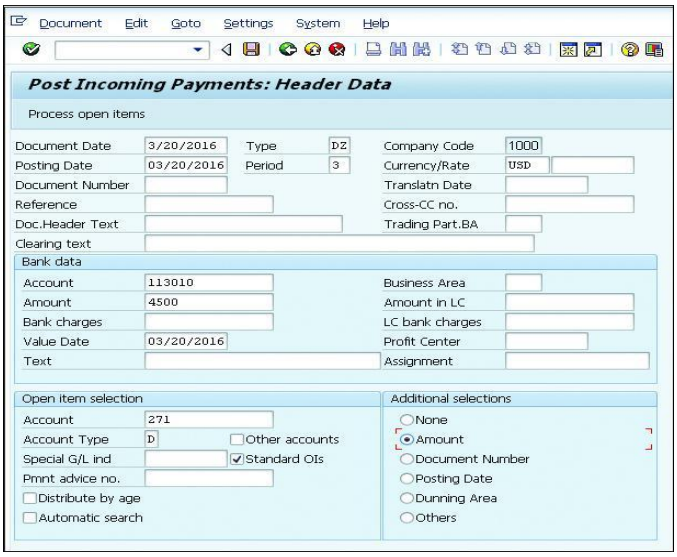

Figure 4: Payment Receipt and AR Clearing — T-Codes F-28 / F-32

7. Transaction Code Summary

T-Code	Transaction	Business Purpose
XD01	Customer Master Create	Onboard customer with SD/FI data
MM01	Material Master Create	Define product with pricing & MRP config
VA11 / VA21	Inquiry / Quotation	Capture interest; issue formal price offer
VA01	Sales Order Create	Book order; trigger pricing & credit check
VA02	Sales Order Change	Modify quantity, dates, or conditions
VL01N	Outbound Delivery	Initiate warehouse picking & dispatch
VL02N	Delivery / PGI	Confirm picking; post stock reduction & COGS
VF01	Billing Document	Generate invoice; auto-create FI accounting entry
F-28	Post Incoming Payment	Record customer payment in bank
F-32	Clear Open Items	Match payment against open AR invoice

Table 2: Key SAP Transaction Codes — O2C Process Reference

8. Process Workflow Overview

The O2C process is a document chain — each step generates a SAP document that the next step references, creating an unbroken audit trail from inquiry to payment:

Inquiry (VA11) → Quotation (VA21) → Sales Order + Credit/Price Check (VA01) → ATP Confirmation → Outbound Delivery (VL01N) → Picking & PGI (VL02N) → Billing/Invoice + FI Posting (VF01) → Payment (F-28) → AR Clearing (F-32) → Cycle Complete

Every module handoff — SD to MM at PGI, SD to FI at billing — is system-driven. No re-entry, no email handoff, no reconciliation lag. This is where the real operational value of SAP lies: not in any individual step, but in the seamless data flow between them.

Figure 5: Complete O2C Workflow — SAP SD Integrated Process Flow

9. Key Benefits

- Faster billing cycle: Invoices are raised on the day of PGI — not days later — directly compressing DSO and freeing up working capital.

- Proactive credit control: Credit checks at order entry catch over-limit customers before goods are dispatched, not after.
- Live inventory accuracy: ATP and PGI together keep system stock in sync with physical stock — no ghost inventory or surprise shortfalls.
- Instant revenue recognition: Accounting entries post automatically at billing — no month-end manual corrections.
- End-to-end traceability: Every document in the chain is linked; any invoice can be traced back to its originating customer inquiry in seconds.
- Single source of truth: Sales, warehouse, and finance work from the same live data — eliminating version conflicts and cross-department delays.

10. Future Enhancements

- SAP Fiori: Replace the classic GUI with mobile-ready Fiori apps (Manage Sales Orders, Track Deliveries) so field teams can act without backend access.
- AI Demand Forecasting: Use SAP Analytics Cloud with embedded ML to predict order volumes by segment, enabling proactive replenishment rather than reactive stock chasing.
- Automated Credit Review: Trigger periodic credit limit reassessments based on payment history — reliable payers rewarded automatically, delinquent accounts flagged.
- Dynamic Pricing Engine: Extend condition techniques with time-based pricing rules, volume tiers, and customer-segment discounts that adjust without manual reconfiguration.
- O2C KPI Dashboard: Deploy real-time SAP Analytics Cloud dashboards monitoring Order-to-Invoice time, DSO, on-time delivery rate, and revenue by sales org.
- SAP Sales Cloud Integration: Link O2C to CRM so leads flow automatically into quotations and won opportunities trigger SD sales orders — closing the loop from pipeline to cash.

11. Conclusion

This project delivers a complete, end-to-end implementation of the Order-to-Cash process in SAP SD for Arjun Industrial Supplies Pvt. Ltd. — covering organizational setup, master data configuration, and the full transactional cycle from customer inquiry through payment clearing.

The integration between SD, MM, and FI is what makes this implementation more than a collection of transactions. It is a synchronized business system where a single customer order triggers coordinated updates across commercial, operational, and financial records — automatically and in real time. The result: faster cash collection, fewer errors, cleaner audits, and a scalable foundation for future growth.

Planned enhancements through SAP Fiori, AI-driven analytics, and CRM integration will further close the gap between customer intent and revenue realization — ensuring the O2C process grows with the business rather than constraining it.